Online Appendix

Fewer Clicks, Not Fewer Searches: Generative AI and the Monetization of Search

Appendix A. Proofs

Let k ≡ γ/(2t), so x̂ = 1/2 + k(a_E − a_I). The AI profits are π_I = s·x̂·(m_I a_I − c) and π_E = s·(1 − x̂)·(m_E a_E − c).

Proof of Proposition 1. For the incumbent,

$$\frac{\partial \pi_I}{\partial a_I} = s[m_I \hat{x} - k(m_I a_I - c)], \qquad \frac{\partial^2 \pi_I}{\partial a_I^2} = -2skm_I < 0$$

At a_I = 0, profit is −s·x̂·c < 0 and the derivative is s(m_I x̂ + kc) > 0. A small increase in a_I therefore strictly raises profit, so a_I = 0 is not a best response. The argument for E is symmetric.

∎

Proof of Proposition 2. Strict concavity implies that best responses are given by the first-order conditions:

$$a_I = \frac{1}{4k} + \frac{a_E}{2} + \frac{c}{2m_I}, \qquad a_E = \frac{1}{4k} + \frac{a_I}{2} + \frac{c}{2m_E}$$

Each best response has slope 1/2, so the best-response map is a contraction and the equilibrium is unique. Substituting the second condition into the first gives (3/4)a_I = 3/(8k) + c/(2m_I) + c/(4m_E). Using 1/(2k) = t/γ yields the stated a_I^e; a_E^e follows by symmetry. Then a_E^e − a_I^e = (c/3)(1/m_E − 1/m_I), which gives x̂^e.

From the incumbent's first-order condition, m_I a_I − c = m_I x̂/k = 2t·m_I·x̂/γ. Hence π_I = s·x̂·(2t·m_I·x̂/γ) = s·(2t·m_I/γ)·x̂².

a. With m_I = m_E = m, a^e = t/γ + c/m; the comparative statics are immediate. (b) The margin is m·a^e − c = m·t/γ and x̂^e = 1/2, so profit is s·m·t/(2γ), independent of c; and

$\partial(m \cdot a^e)/\partial c = 1$. (c) If $m_I > m_E$, then $a_I^e - a_E^e = (c/3)(1/m_I - 1/m_E) < 0$ and $\hat{x}^e > 1/2$. Interiority requires $(\gamma c/(6t)) \cdot |1/m_E - 1/m_I| < 1/2$. ∎

Proof of Proposition 3. With $a_E = 0$ and $m_I = m$, the incumbent's best response is $a_I = 1/(4k) + c/(2m) = t/(2\gamma) + c/(2m)$. Then $\hat{x} = 1/2 - k \cdot a_I = 1/4 - \gamma c/(4tm)$, which is positive if and only if $m \cdot t > \gamma \cdot c$. The margin is $m \cdot a_I - c = (m \cdot t - \gamma \cdot c)/(2\gamma)$, and $\hat{x} = (m \cdot t - \gamma \cdot c)/(4tm)$. Multiplying gives $\pi_I = s(m\,t - \gamma\,c)^2/(8\gamma\,m\,t)$. Dividing by the symmetric-equilibrium profit $s \cdot m \cdot t/(2\gamma)$ gives $(1/4)(1 - \gamma c/(mt))^2 \le 1/4$. The entrant's profit is $-s(1 - \hat{x})c < 0$. ∎

Proof of Proposition 4. Equilibrium ad intensities do not depend on s, because s multiplies both AI profit functions. Write $\pi_I^{AI} = s \cdot \varphi$, where φ is the incumbent's equilibrium AI profit per AI-amenable query. Then $\Pi_I = (1 - s)R + s\varphi$ and $d\Pi_I/ds = \varphi - R$. (i) With $a_I = 0$, $\varphi = -\hat{x}c < 0$, so $d\Pi_I/ds < -R < 0$ and $\Pi_I \to -\hat{x}c < 0$ as $s \to 1$. (ii) In the symmetric equilibrium $\varphi = m \cdot t/(2\gamma)$, so $d\Pi_I/ds \ge 0$ if and only if $m \ge 2\gamma R/t$. ∎

Proof of Proposition 5. Each platform takes m as given, so the first-order conditions of Proposition 2 apply with $m_I = m_E = m$. In the symmetric equilibrium, $a = t/\gamma + c/m$ and $\hat{x} = 1/2$, so $A = (1 - s)\bar{a} + s(t/\gamma + c/m)$. Substituting into $m = \mu - \eta A$ and multiplying by m gives $m^2 - Bm + \eta sc = 0$. Real roots exist if $B^2 \ge 4\eta sc$. The larger root is the stable one: there, $2m - B = \sqrt{(B^2 - 4\eta sc)} > 0$, which is equivalent to $1 - \eta sc/m^2 > 0$. Differentiating the quadratic in c gives $(2m - B) \cdot dm/dc = -\eta s$, hence $dm/dc = -\eta s/\sqrt{(B^2 - 4\eta sc)} < 0$. Incumbent AI profit $s \cdot m \cdot t/(2\gamma)$ and legacy revenue $(1 - s)m\bar{a}$ are increasing in m and therefore decrease in c. ∎

Proof of Proposition 6. Holding a_I fixed, differentiate Π_I with respect to a_E. Through $\hat{x}$, the AI term changes by $s \cdot k \cdot (m\,a_I - c)$. Through m, the legacy and AI terms change by $[(1 - s)\bar{a} + s\,\hat{x}\,a_I] \cdot \partial m/\partial a_E = V_I \cdot \partial m/\partial a_E$, where $\partial m/\partial a_E = -\eta \cdot \partial A/\partial a_E = -\eta s[(1 - \hat{x}) + k(a_I - a_E)]$. The incumbent's first-order condition (with m given) is $k(m\,a_I - c) = m\,\hat{x}$. Substituting yields the stated expression. At the symmetric equilibrium, $\hat{x} = 1/2$ and $a_I = a_E$, so $\partial\Pi_I/\partial a_E = (s/2)$

$(m - \eta V_I)$. This is positive if and only if $\eta V_I/m < 1$. Writing $\eta V_I/m = (\eta A/m)(V_I/A) = \varepsilon w_I$ gives the condition. ∎

Proof of Proposition 7. In the symmetric equilibrium, the incumbent's AI margin is $m \cdot a - c = m \cdot t/\gamma$ and its share is 1/2, so $\Pi_I = (1 - s)m\bar{a} + s \cdot m \cdot t/(2\gamma) = m\Lambda$. Then $d\Pi_I/ds = m \cdot \Lambda'(s) + \Lambda \cdot m'(s)$, with $\Lambda'(s) = t/(2\gamma) - \bar{a}$. Differentiating $m = \mu - \eta[(1 - s)\bar{a} + s(t/\gamma + c/m)]$ with respect to s gives $m' = \eta(\bar{a} - a) + \eta sc \cdot m'/m^2$, so $m' = \eta(\bar{a} - a)/(1 - \eta sc/m^2)$. The denominator is positive at the stable root (Proposition 5), so m' has the sign of $\bar{a} - a$. With $\eta = 0$, the condition $d\Pi_I/ds \geq 0$ reduces to $t/(2\gamma) \geq \bar{a}$, that is, $m \cdot t/(2\gamma) \geq m\bar{a} = R$, as in Proposition 4. ∎

Proof of the Lemma in Section 4.6. With the incumbent's format fixed, $\hat{x} = 1/2 + (\gamma a_E - g)/(2t)$, and the entrant maximizes $s(1 - \hat{x})(m\, a_E - c)$, which is strictly concave in a_E. The first-order condition is $m(1 - \hat{x}) = k(m\, a_E - c)$ with $k = \gamma/(2t)$. Substituting $1 - \hat{x} = 1/2 - k\, a_E + g/(2t)$ gives $a_E = (t + g)/(2\gamma) + c/(2m)$. Then $\gamma a_E = (t + g)/2 + \gamma c/(2m)$, and $\hat{x} = 1/2 + [(t + g)/2 + \gamma c/(2m) - g]/(2t) = (3t - g)/(4t) + \gamma c/(4tm)$. The incumbent earns r on each of its $\hat{x}$ queries. ∎

Proof of Proposition 8. (i) Summaries dominate links if $(R - c_O)\hat{x}(\Delta - \Delta_O) > R \cdot \hat{x}(\Delta)$. Multiplying by 4t and rearranging gives $R\Delta_O > c_O(3t - \Delta + \Delta_O + \gamma c/m)$. (ii) Full answers dominate summaries if $m \cdot t/(2\gamma) \geq (R - c_O) \cdot [(3t - \Delta_H)/(4t) + \gamma c/(4tm)]$. Multiplying by $4t/(R - c_O)$ gives $2mt^2/(\gamma(R - c_O)) \geq 3t - \Delta_H + \gamma c/m$, which rearranges to $\Delta_H \geq \Delta_H^*$. (iii) $\partial\Delta_H^*/\partial R = 2mt^2/(\gamma(R - c_O)^2) > 0$; $\partial\Delta_H^*/\partial c = \gamma/m > 0$; $\partial\Delta_H^*/\partial m = -\gamma c/m^2 - 2t^2/(\gamma(R - c_O)) < 0$; $\partial\Delta_H^*/\partial c_O = -2mt^2/(\gamma(R - c_O)^2) < 0$. The corollary follows because Δ_H^* is finite while $m \cdot t/(2\gamma) < R$ is a separate restriction on parameters; both can hold at once (for example, $t = \gamma = 1$, $m = 2$, $c = 0.3$, $R = 1.4$, $c_O = 0.05$ gives $\Delta_H^* \approx 0.19$ and $m \cdot t/(2\gamma) = 1 < R$). ∎

Proof of Proposition 9. Fix a date d and let $h(\sigma, d) = \sigma[\pi_A - (R_0 - c_O)\hat{x}_H(d)] - R_1\hat{x}_H(d)\sigma^2$. Since $R_1\hat{x}_H(d) > 0$, h is strictly concave in σ, so the set $S(d) = \{\sigma : h(\sigma, d) \geq K\}$ is an interval, possibly empty. Moreover $\partial h/\partial d = -\hat{x}_H'(d)[(R_0 - c_O)\sigma + R_1\sigma^2] > 0$ for $\sigma > 0$, because $\hat{x}_H' < 0$ and $R_0 > c_O$; hence $S(d) \subseteq S(d')$ for $d < d'$. The lower contour sets of the

switching date, $\{\sigma : D(\sigma, K) \leq d\} = S(d)$, are therefore nested intervals, which is the definition of quasi-convexity of D in σ and gives part (i). For part (ii), a higher K shrinks S(d) at every d, so D weakly rises. Part (iii) follows because an engine outside every S(d) never satisfies the switching condition, and a fall in K enlarges every S(d). ∎

Proof of Proposition 10. G is continuous and decreasing in θ, and T is continuous and decreasing in ω, so $\Phi = T \circ G$ is continuous and increasing. Since $\omega \in [0, 1]$, Φ maps $[q, \bar{\theta}]$ into itself, where $\bar{\theta} = 1 - (1 - q)\zeta$, and by Brouwer's theorem (or Tarski's, since Φ is increasing) a fixed point exists. (i) At a fixed point, $\theta = 1 - (1 - q)[\zeta + (1 - \zeta)\omega]$, so $\theta - q = (1 - q)(1 - \zeta)(1 - \omega) \geq 0$, decreasing in ω. (ii) At the lower end of the interval $\Phi(q) \geq q$, and at the upper end $\Phi(\bar{\theta}) \leq \bar{\theta}$. If Φ crosses the 45-degree line from below at an interior fixed point θ_2, continuity implies at least one fixed point in $[q, \theta_2)$ where Φ crosses from above and one in $(\theta_2, \bar{\theta}]$. Under the adjustment $\theta_{n+1} = \Phi(\theta_n)$, fixed points where Φ crosses from above are stable and θ_2 is unstable; the lowest has the most clicks (vigilant) and the highest the fewest (complacent). (iii) If $e/(1 - q)$ exceeds the upper bound of the support of F, then $G(\theta) = 0$ for all $\theta \geq q$, so Φ is constant at $\bar{\theta}$, its unique fixed point. (iv) Φ is decreasing in ζ at every θ. For ζ large enough, Φ lies below the 45-degree line on $(\theta_2, \bar{\theta}]$, so the monotone sequence θ_n falls below θ_2 in finite time. When ζ returns to its original value, θ_n lies in the basin of the vigilant equilibrium and converges to it. ∎

Appendix B. Data and Replication Package

All analyses use publicly available data. The replication package (replication_package.zip) contains the data-collection scripts, the analysis scripts that produce each table and figure, the numerical examples for Propositions 8–10, small hand-coded datasets (event dates, rollout dates by edition and country, launch dates and revenue for 17 search engines), and the processed data for the trust-shock analysis. Its README maps every script to the corresponding table or figure and lists data sources, access dates, run order, and software requirements.

Raw data that third parties do not permit us to redistribute (for example, Google Trends extracts and Wikimedia hourly dump files) are re-downloaded by the collection scripts. Google Trends is rate-limited and returns sampled indices, so re-downloaded series can differ slightly from ours.

Data sources. Wikimedia hourly projectviews dumps (dumps.wikimedia.org); Google Trends (via pytrends); Eurostat datasets isoc_ai_iaiu and isoc_ci_ifp_fu; World Bank World Development Indicators (NY.GDP.PCAP.CD, IT.NET.USER.ZS, SP.POP.TOTL); StatCounter Global Stats; Alphabet, Meta, and Microsoft filings on SEC EDGAR (Forms 8-K, 10-Q, XBRL); daily stock prices for the event study; and launch dates from company announcements, archived help pages, and trade press as cited in the text. All sources were accessed in October 2026.

Figure data. The values plotted in Figures 1–5 are provided in figures/figure_data/ together with the plotting script (make_figures.py).